



The vortex-induced vibrations of a triple-box girder model of a long-span rail-cum-road bridge

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ABSTRACT

Triple-box girders, i.e., three separated box girders are arranged in tandem, are gradually used in super-span bridges because of their excellent flutter stability. However, the gaps between triple boxes can lead to complex vortex dynamics around the structures that result in vortex-induced vibrations (VIVs). In the present study, we experimentally investigate VIVs in a long-span bridge with a triple-box girder cross-section using wind tunnel tests. In the static test, in which the test model is fixed, a ‘dual-frequency’ phenomenon that expands the VIV lock-in region is observed in the global flow field. Furthermore, the results of a dynamic test indicate that the existence of gaps is essential to the occurrence of VIVs in triple-box girders. The smoke-wire flow visualization technique is employed to explore the excitation mechanism by examining the surrounding flow dynamics of the triple-box model. Separated motion-induced vortices (MIVs) and von Karman vortices (KVs) are found to be coupled with structural vibrations to sustain the VIVs. Five stages of VIV regions, along with five types of vortices in the surrounding flow field, are identified for the girder model. Finally, the effects of accessory structures of the bridge model on the VIV responses are also studied. This study contributes to our understanding of the unsteady flow dynamics and VIV mechanisms of a typical triple-box girder.

1. Introduction

Societal and economic development has boosted the design and construction of super-long bridges. To meet the increasing demands for greater bridge spans, separated box girders have been widely adopted for long-span bridges because of its excellent flutter stability and the limited span of single-box girders. Through the use of a twin-box girder and a triple-box girder, the main spans of the Gibraltar Bridge and the Messina Bridge respectively reached 3550 m and 3300 m (Diana et al., 1999; Larsen, 1998). Some other famous bridges have also employed separated-type configurations, including the Xihoumen suspension bridge (main span: 1650 m), the Gwangyang bridge (main span: 1545 m), and the Stonecutter bridge (main span: 1018 m), although they are shorter than the single-box-girder limit. The motivation behind this is that flutter stability can be effectively improved with slots at the center of a bridge girder (Sato et al., 2000; Baldomir et al., 2013). Moreover, a triple-box girder is superior to single- and twin-box configurations in terms of installation, maintenance, freedom of pylon design, and transversal stiffness (Wang and Saul, 2020). However, increasing the main span of a structure leads to higher flexibility and a lower damping ratio. For example, the frequency of the Xihoumen suspension bridge

(with a main span of 1650 m) is 0.098 Hz (Li et al., 2011). Consequently, the flexible bridge girder is quite sensitive to wind loads, and under certain conditions, complex structure-fluid coupling problems such as amplitude-limited vortex-induced vibrations (VIVs) (Li et al., 2021) and disastrous flutter instabilities may develop. Meanwhile, with the existence of gaps, the flow dynamics become more complicated and VIV problems can be exacerbated (Larsen, 2005).

Many researchers have studied VIVs through field monitoring, wind tunnel tests, numerical simulations, and theoretical analysis (Gao et al., 2021). VIVs are wind-induced vibrations that have been observed in a large number of bridges (e.g., Frandsen, 2001; Fujino and Yoshida, 2002; Larsen et al., 2000; Li et al., 2014, 2011). For example, Larsen et al. (2000) observed vertical oscillations of the Storebælt bridge and attributed these oscillations to periodic von Karman vortex (KV) shedding from the main girder. Generally, it was seen that when the inflow passed the girder, KVs were shed from the trailing edge if the Reynolds number was beyond a certain threshold. The dominant shedding frequency satisfied the Strouhal law ($St = fD/U$, where D and U represent the characteristic length and the inflow speed). It was observed in early field monitoring that when the vortex shedding frequency approached some order of the structural frequency, it would be locked at the

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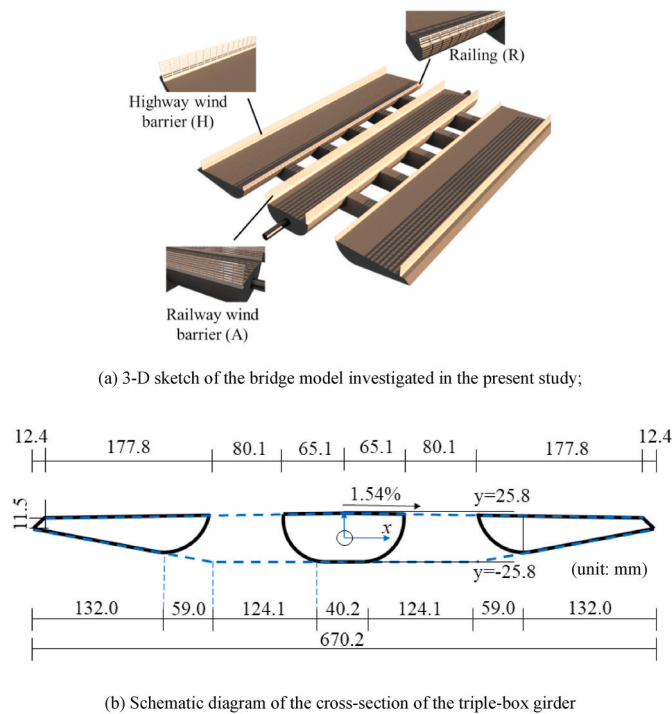


Fig. 1. Schematic of the experiment model in the present study.

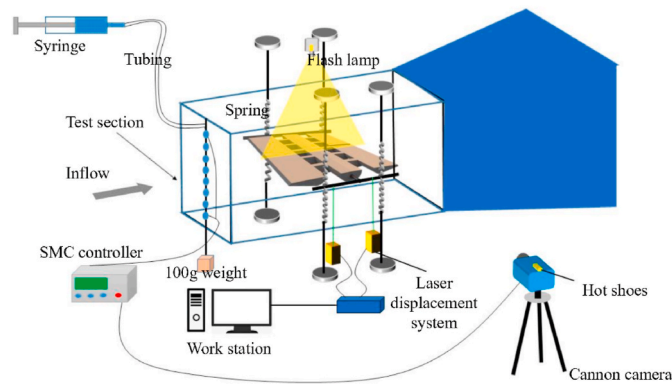


Fig. 2. Experimental layout of the present study.

Table 1

The dynamic parameters of the prototype and the model.

Parameter	Unit	Prototype	Bare girder model	Completed bridge model
Length (L)	m	/	0.750	0.750
Width (B)	m	67.020	0.670	0.670
Height (D)	m	5.160	0.051	0.051
Equivalent mass (m)	kg/ m	74 937.00	8.11	8.11
Vertical damping ratio (ξ_v)	%	0.30	0.11	0.11
Vertical natural frequency (f_v)	Hz	0.1359	10.25	10.13

structural frequency in a certain wind speed region (i.e., the ‘lock-in’ phenomenon) (Frandsen, 2001). The lock-in phenomenon is one of the most important characteristics of VIVs (Behara et al., 2017; Kumar et al., 2016; Liu and Jaiman, 2016). In addition to the well-known KVs, previous studies have revealed that motion-induced vortices (MIVs) related to the enhancement of shear layer instabilities could induce VIVs as

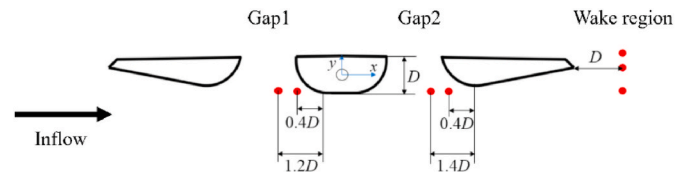


Fig. 3. Locations of the hotwire measured positions.

we4ll (Matsumoto et al., 1993; Shiraishi and Matsumoto, 1983). For bridge box sections, the slenderness ratio B/D (where B and D are the chord length and box height, respectively) is usually greater than 3. According to Matsumoto et al. (1993), MIVs and KVs can exist simultaneously and interact in a complicated way. These interactions are strongly related to the geometric shapes of bridge box sections. However, the target research object of Matsumoto et al. (1993) was a 2D rectangular cylinder. When the concepts of KVs and MIVs are introduced to explain the VIV mechanisms of multiple-box structures, i.e., a twin-box bridge girder or a triple-box bridge girder, the KV-spoiling effect of the downstream box must be taken into consideration. The VIV amplitude of the upstream box girder increased due to the KV-spoiling effect of the downstream one (Morikawa et al., 1991). In fact, the KV-spoiling effect on the VIV was complex and both KVs and MIVs could be affected.

Li et al. (2014, 2011) monitored the wind field and wind-induced responses of the twin-box girder of the Xihoumen Suspension bridge. Later, Wu and Laima (2021) further investigated the surrounding flow field of the twin-box girder when a VIV occurred using smoke-wire flow visualizations. Before the VIV, vortices related to the Kelvin–Helmholtz (K–H) instability were shed from the upper and lower surfaces of the stationary girder and merged in its wake. No interactions between the shear layers could be seen in the gap. When the VIV happened, alternative vortex shedding appeared in the gap. Meanwhile, the KV through the upstream box girder was spoiled by the existence of the downstream one, and continued to propagate downstream in the form of large-scale discrete vortices. The KV was shed downstream in the wake. The flow dynamics coupled with the VIV of the twin-box girder represent a classic case, containing the interactions of KVs and MIVs and the KV-spoiling effect simultaneously. A thorough understanding of the VIV mechanism is needed to improve control methods (Chen et al., 2019; Yang et al., 2020; Yang and Chen, 2021).

The aerodynamical and aeroelastic performance of twin-box girders has been investigated in detail (Ma et al., 2020; Tang et al., 2019; Wu and Laima, 2021; Yuan et al., 2017). Nevertheless, research on triple-box girders has only begun in China in recent years to meet engineering requirements, including those of a suspension bridge in the southeast of China (main span: 1488 m, under construction), the Taoyaomen rail-cum-road Bridge (main span: 666 m, under construction), and two bridges with triple-box girders that are in the design stage (Wang et al., 2021). Due to the existence of the two gaps in twin-box girders, their aerodynamics and VIV responses are complex (Diana et al., 2006, 2013; Meng et al., 2022). Brancaloni and Diana (1993) showed that deck optimization could significantly improve aerodynamic stability in such situations. Wang et al. (2011) studied different aerodynamic configurations with accessory structures in a wind tunnel test. Their results showed that a 15° inclined web could increase the flutter critical wind speed and diminish VIVs. Wang et al. (2021) studied the effects of linear and curved webs on the VIV performance of the Taoyaomen rail-cum-road Bridge via a wind tunnel test. Wang et al. (2022) revealed that the shear layers through the leeward side of the windward and middle boxes impinged on the windward side of the downstream boxes. This impinging-shear-layers effect caused significant pressure fluctuations there that were likely to induce VIVs. The target bridge of Wang et al. (2022) was designed with linear webs and more attention was paid to VIV control. Yang et al. (2021a) numerically studied the VIV mechanism of a completed triple-box girder with curved

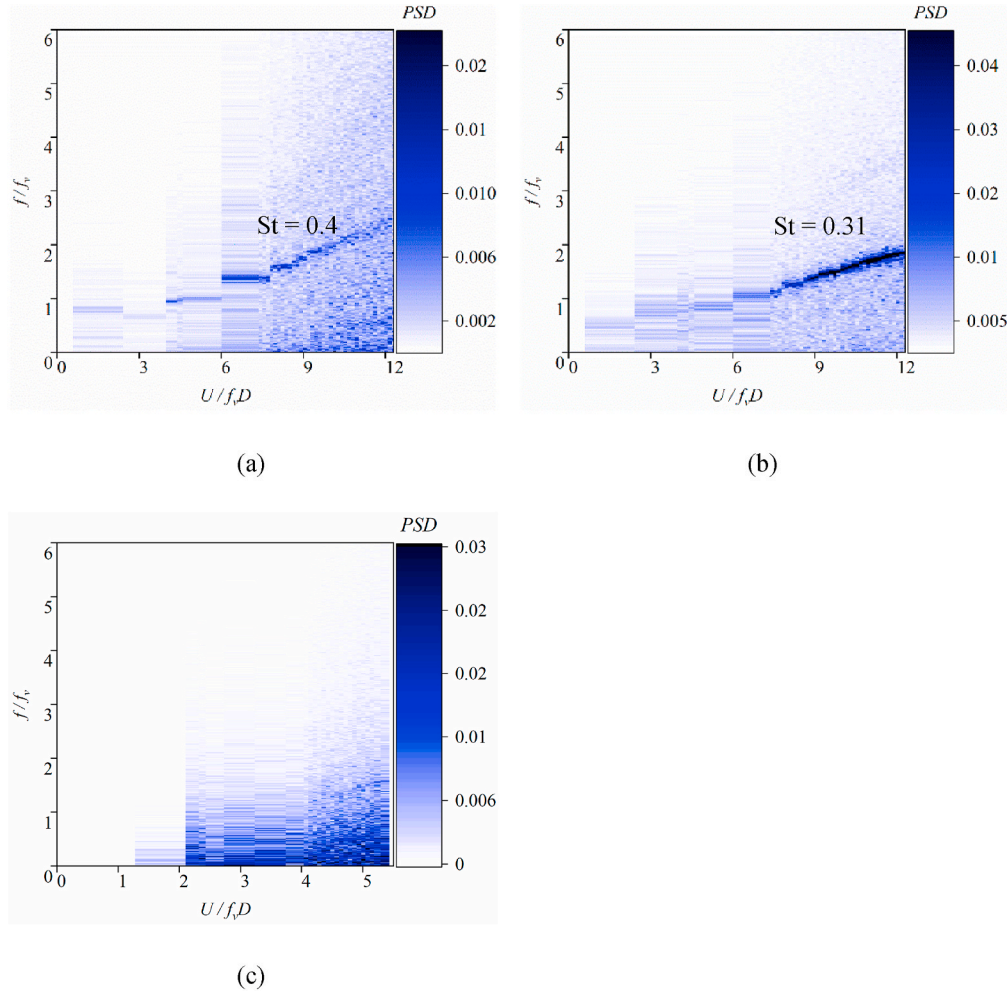


Fig. 4. PSD heat map for various locations of the static bare girder. (a) Gap1; (b) Gap2; (c) Wake.

Table 2

Test cases for the bare girder.

Cases	Gaps covered	Possible source of excitation
Case 0	None	Gap1+Gap2+wake
Case I	Gap2	Gap1+wake
Case II	Gap1	Gap2+wake
Case III	Gap1+Gap2	wake

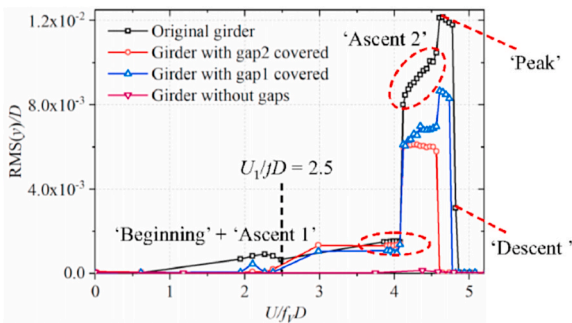


Fig. 5. Transverse displacement of the bare girder with various gaps covered.

webs. It was revealed that the large-scale vortex shedding formed behind the safety barrier of the upstream box and the maintenance tracks, above the downstream highway box, should be responsible for severe VIVs.

The torsional VIV (T-VIV) of the same triple-box girder was reproduced in a wind tunnel (Yang et al., 2021b) and its mechanism was analyzed by vortex-excited force (VEF) evolutionary features during the VIV process. However, the vertical VIV (V-VIV) mechanism of a bare girder with curved webs has not yet been investigated.

The VIVs of a long-span bridge extract vibrational energy from the fluid-structure (wind-bridge) system. Therefore, the surrounding flows and vortical structures play an important role in understanding the VIVs of a triple-box girder. Despite the rapidly rising number of studies on the VIV behaviors of triple-box girders of long-span bridges, more investigation of the mechanisms behind them is needed from the perspective of vortex dynamics and flow structures. Therefore, the present study experimentally investigates the aeroelastic characteristics and flow dynamics of a triple-box girder model. The response and mechanism of the VIVs are discussed and the effects of accessory structures are studied via wind-tunnel tests. This paper is arranged as follows. The experimental setup is described in Sec. 2. The experimental data are analyzed and discussed in Sec. 3. The conclusions are provided in Sec. 4.

2. Experimental setup

The experimental study was performed in the closed-loop wind tunnel (SMC-WT2) at the Harbin Institute of Technology. The dimension of the test section was 800 mm (width) $\times$ 1200 mm (depth) $\times$ 3000 mm (length). The maximum wind speed was 20 m/s while the minimum was about 2 m/s with a very low turbulence intensity ($<0.4\%$).

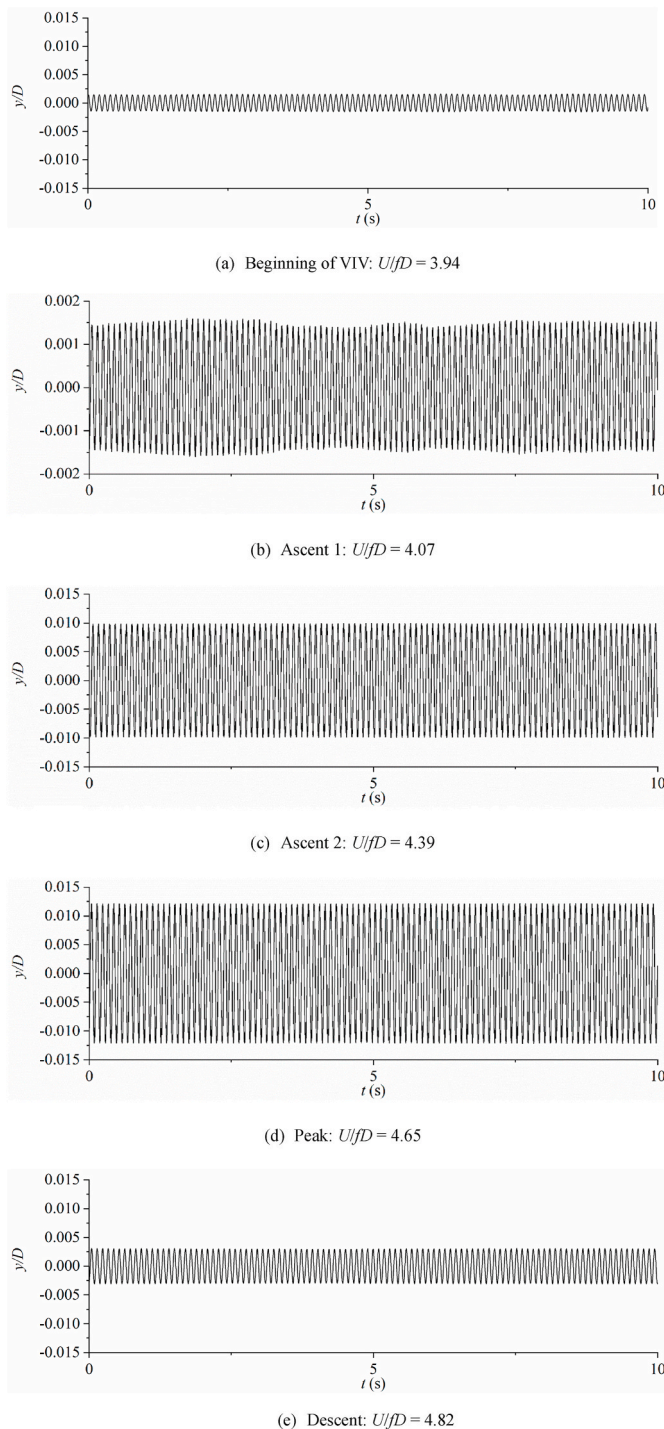


Fig. 6. Time histories of vertical displacement at various VIV stages.

2.1. Bridge prototype and the test model

The prototype of the present model was a long-span rail-cum-road bridge in the southeast of China, which was under construction at the time of writing. The bridge pylon was 294 m in height. The total span was 2664 m with a main span of 1488 m. The cross-section was composed of triple-separated streamlined boxes, with a central box of 5.16 m in height, as shown in Fig. 1(a).

The test model was on a geometrical scale of 1:100 and the girder slope was equal to that of the prototype. Fig. 1(a) shows a 3D sketch of the completed bridge model. The bare girder and the accessory

structures were made of acrylonitrile-butadiene-styrene (ABS) engineering resin. The triple-separated boxes were connected by five pairs of coupling beams. A schematic diagram of the cross-section is shown in Fig. 1(b). The span length (B) of the test model was 670.2 mm and the height (D) was 51.6 mm. The blockage ratio was calculated as 4.03% and the solid blockage effect was considered negligible in the present study. To study the VIV mechanism, the gap was covered along the blue dotted lines for various cases in Fig. 1(b). Three kinds of accessory structures were considered for the completed bridge girder: a highway wind barrier (abbreviated H), a railing (R), and a railway wind barrier (A). Highway and railway wind barriers kept the same porous rate as the original bridge. The railing was designed with a geometrical scale of 1:100. Unless stated otherwise, the angle of attack (AOA) in the present experiments was kept at 0° .

A threaded rod ran through the central box to create a spring-suspended sectional model (SSSM) system in the wind tunnel, as shown in Fig. 2. The SSSM system, which was installed on the rigid frames outside the test section, was designed to study the dynamic characteristics of the bridge model as well as the dynamic flows around it. The test model was supported by four pairs of springs and the model could only vibrate vertically. A free vibration test on the SSSM system was conducted initially in the still air to obtain the natural frequency and the damping ratio, both for the bare girder and the completed bridge model, and these values are listed in Table 1. The vertical natural frequency of the completed bridge model was 10.13 Hz, which was slightly smaller than the bare one (10.25 Hz). According to Table 1, the Scruton number, ($Sc = 2\pi M\xi/\rho D^2$) is 43.3 for the real bridge and 17.6 for the present test model, where M is the unit mass, ξ is the damping ratio, and ρ is the air density. It should be noted that the vertical VIV amplitude of a box girder decreased as the Sc number increased, as reported in Wu and Laima (2021). Therefore, the relatively lower value of Sc number chosen for the model in the present study would increase the possibility of exciting VIV and amplify the potential VIV amplitude.

2.2. Measurement apparatus

Two non-contact laser displacement sensors (Keyence, LK-G500, Japan) were used to measure the dynamic responses of the test model. The sampling frequency was 1000 Hz, the absolute accuracy was within 0.05%, and the measurement range was less than 500 mm. To obtain the fluctuating flow velocities at certain crucial and sensitive points, hot-wire anemometers were employed and placed at different locations. As shown in red in Fig. 3, the hot-wire measurements were taken at the gaps and in the wake of the triple-box girder. The sampling frequency was set to 1000 Hz. The VIV responses and the flow velocities were collected 120 s after the wind speed was adjusted, as the test model vibrated steadily.

Smoke-wire flow visualization was also performed to characterize the dynamic flows around the model when the VIV amplitude reached a maximum, as shown in Fig. 2. The visualization system included the following parts: a digital single-lens-reflex (SLR) Canon 5D Mark II camera, a flash lamp, a 0.1-mm resistance wire, and a control system (Hanghua SW02 smoke-wire controller). The control system utilized the capacitor discharge to minimize the diameter of the resistance wire under high current, and as a result, the smoke lines became clearer and more hierarchical. Since the capacitor discharge time was very short, the system controlled the camera shutter and the flash as well (where the hot shoes were working). In addition, a syringe was used to add the paraffin oil drops onto the resistance wire in the wind tunnel so that the incoming flow inside was not disturbed.

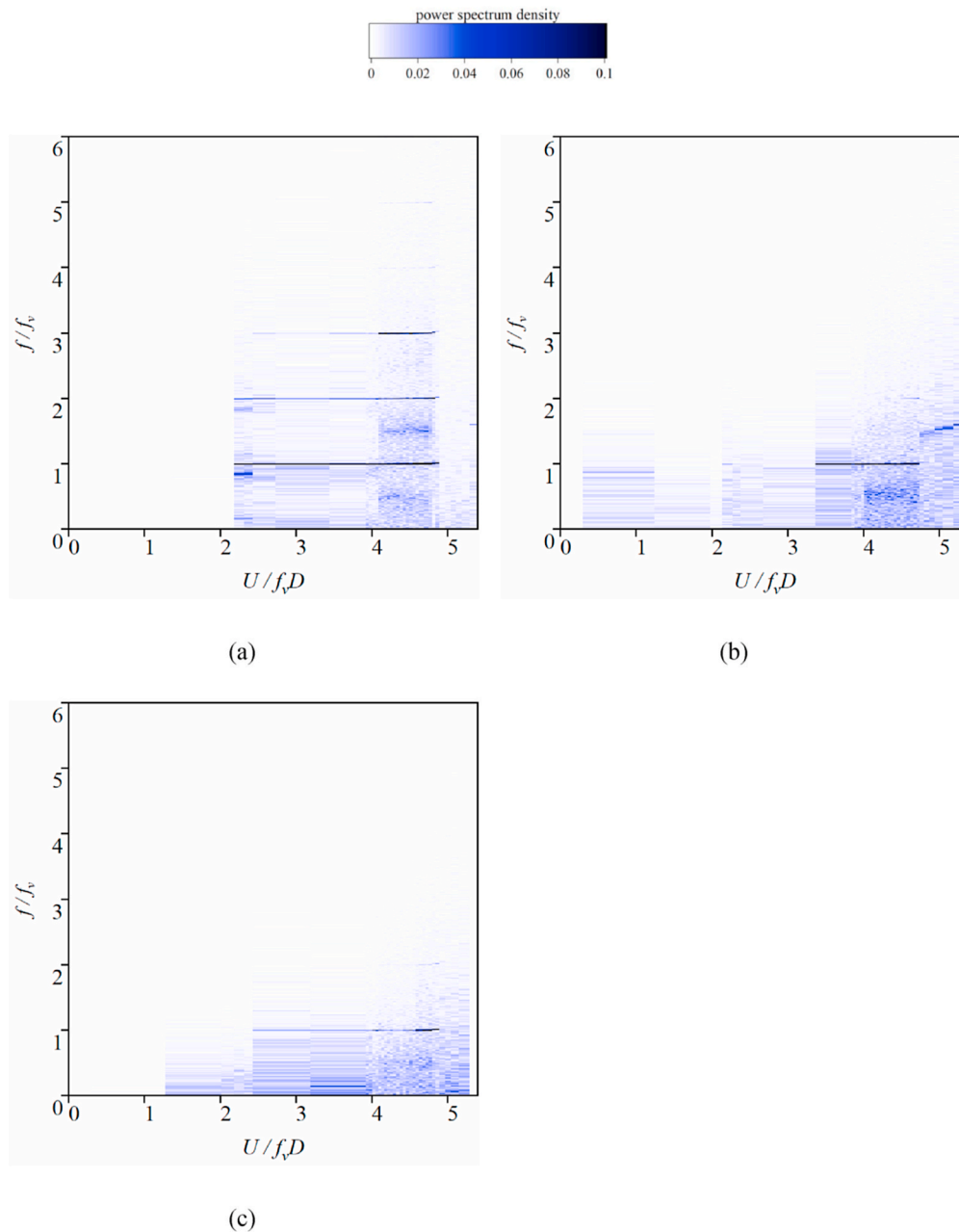


Fig. 7. PSD contours for various locations of the vibrating bare girder (case 0). (a) Gap1; (b) Gap2; (c) Wake.

3. Results and discussion

3.1. The bare girder

3.1.1. Static test

To obtain the Strouhal number (St) of the girder model, we fixed the girder model and measured the point wind velocities in three regions: gap1, gap2, and the wake region. Wind velocities at seven locations were measured, as marked by red dots in Fig. 3. In the gaps, the power spectrum density (PSD) amplitudes close to the box surfaces were larger than those in the gap center. Thus, we show the heat map of the PSD of the normalized wake frequencies for the static bare girder versus the reduced wind velocity ($U/f_v D$) in Fig. 4. The shape of the girder configuration was so streamlined that we could not identify a dominant frequency in the wake. However, we obtained $St_1 = 0.4$ and $St_2 = 0.31$ in gap1 and gap2, respectively. This 'dual-frequency' phenomenon has also been observed in the aerodynamic lift of three tandem circular cylinders

(Alam et al., 2018), and was attributed to secondary vortices, induced by the merging and stretching of primary vortices. Hu et al. (2019) also reported the 'double-vortex model' for a typical bridge girder. The secondary frequency was accompanied by the generation of secondary vortices, and therefore, the appearance of dual frequencies might affect the VIV behaviors of a triple-box girder, as will be discussed below. The St_1 and St_2 in the present study were not harmonic, so there were two kinds of vortices in the overall flow field. The origin of the 'dual-frequency' phenomenon has been explained in other work and will not be explained further here.

3.1.2. VIV behavior of the bare girder

To verify the respective roles of the flows in the gaps and wake in the excitation of VIVs, we utilized the SSSM system to reproduce the VIVs of four cases, as shown in Table 2. The VIV amplitude variations versus the reduced inflow velocities are shown in Fig. 5.

It was found that when all gaps were covered (case III), the VIV of the

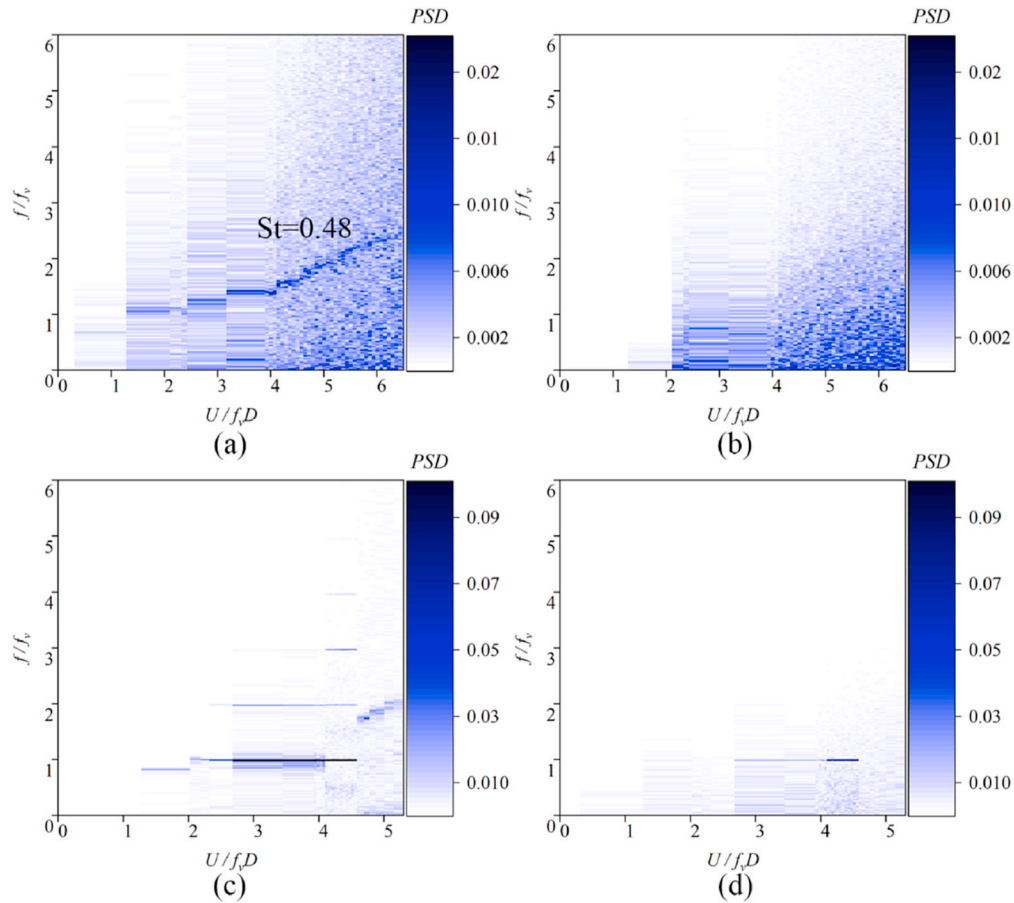


Fig. 8. PSD contours for various locations of case I. (a) Gap1 & (b) Wake of the static model; (c) Gap1 & (d) Wake of the VIV model.

plate-like box girder did not develop, indicating that a VIV cannot be excited by the wake alone and that the existence of gaps is essential to the occurrence of VIVs. As for the original girder (case 0), since St_1 and St_2 were very close, there existed only one VIV region. The VIV amplitude of the original girder was larger than that of the other cases, indicating that the two gaps resulted in more violent and complex VIVs. In addition, the VIVs in case I had smaller amplitudes than those in case II.

From Fig. 5, the development of a VIV can be divided into five stages. In the 'Beginning' region and the 'Ascent 1' region, there exists a platform area, which is considered unsteady. The shape of the girder was so streamlined that it started to show VIVs at a very low wind speed. Fig. 6 shows the time histories of the vertical displacement at various VIV stages. It was verified that the vibrational amplitude of the triple-box girder model became steady in the 'Ascent 2' region.

Fig. 7 shows the PSD contours of the normalized flow frequencies of the vibrating bare girder. The 'lock-in' phenomenon, which is a typical feature of VIVs, was clearly identified. Within the lock-in range, the vortex shedding frequency was captured by the structural natural frequency and no longer followed the Strouhal law. In addition, harmonic frequencies could be found up to five orders in the dynamic flows of gap1. However, this was not obvious in gap2 and the wake flow region.

Figs. 8–10 show the PSD contours of the normalized flow frequencies for cases I–III. As shown in Fig. 8(a) and (b), when gap2 was covered for the static model, the Strouhal number St_1 was 0.48. However, there was no dominant frequency (i.e., periodic vortex shedding) in the wake of the triple-box girder. While gap1 was covered for the original bare girder [Fig. 9(a) and (b)], the Strouhal number St_2 was 0.34 for the unsteady flows in gap2 and St_w was 0.39 in the wake. Thus, the 'dual-frequency' phenomenon occurred. When both gap1 and gap2 were covered (Fig. 10), no dominant wake frequency could be identified. Therefore,

for the present girder, the wake could not excite VIVs alone, as demonstrated in Fig. 5. Figs. 8 and 9 (c) and (d) show the PSD contours of the VIV cases. Multiple harmonic frequencies could be found in the uncovered gap (gap1 for case I and gap2 for case II), which could also potentially excite multiple VIV regions.

3.1.3. Smoke-wire flow visualization

Smoke-wire flow visualization was conducted to reveal the flow dynamics around the triple-box girder. Fig. 11 shows the flow dynamics at various phases around the vibrating bare girder while the VIV amplitude reached a maximum ($U/f_v D = 4.65$). As shown in Fig. 11, four groups of vortices were observed (Groups 1–4).

Group 1 (MIV-1): Due to the heaving motion of the original girder, the inflow separated strongly at the leading edge of the upstream box and then generated separated vortices. The separated vortices were small in scale.

Group 2 (KV-1): With the heaving motion of the original girder, the MIV-1s on both sides through the upstream box were squeezed, then buckled, interacted with each other, and formed an alternative shedding KV in gap1.

Group 3 (MIV-2): Flows through gap1 impinged on the central box. Stronger flow separations then occurred at the leading edge of the central box and large-scale separated vortices appeared. As the girder reached its highest (lowest) point, the upper (lower) large-scale vortex reattached to the upper (lower) surface of the central box.

Group 4 (MIV-3): Similar vortices to those in Group 3 were generated at the leading edge of the downstream box. After separation at the leading edge of the downstream box, the scale of the separated vortices also got larger. The increase in the scale of the separated vortices could possibly be responsible for the enhancement of the VIV.

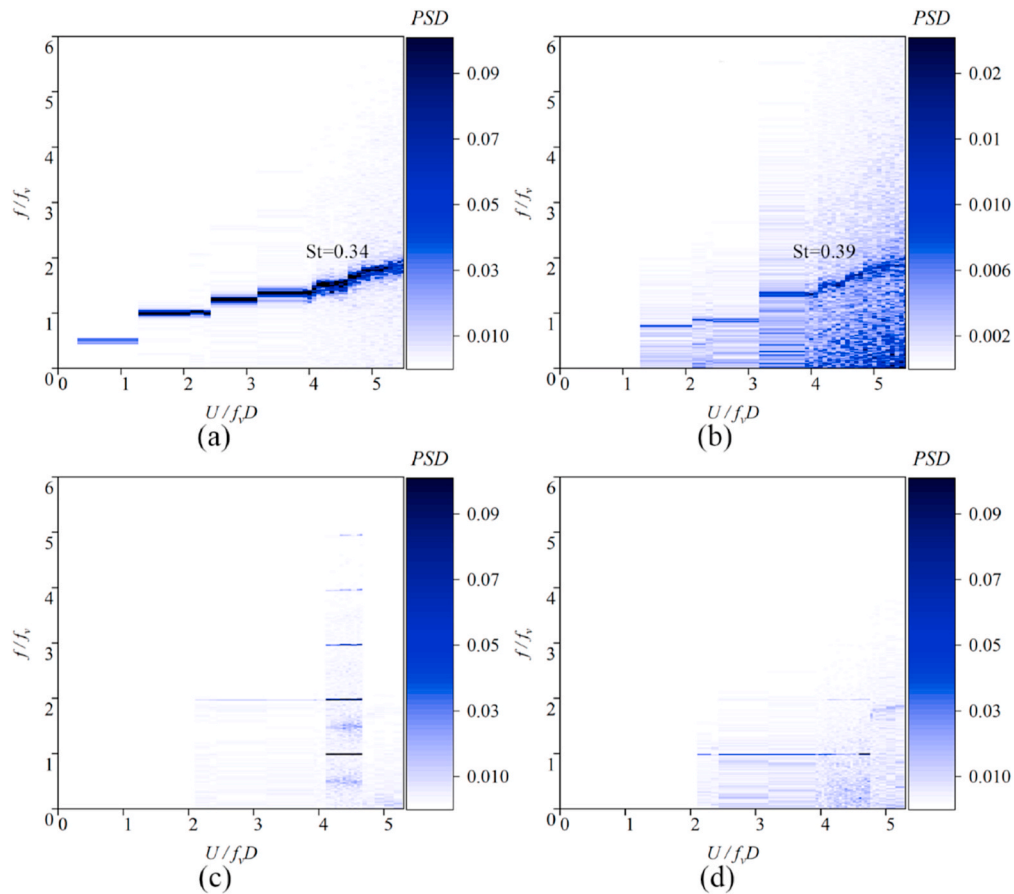


Fig. 9. PSD contours for various locations of case II. (a) Gap2 & (b) wake of the static model; (c) Gap2 & (d) wake of the VIV model.

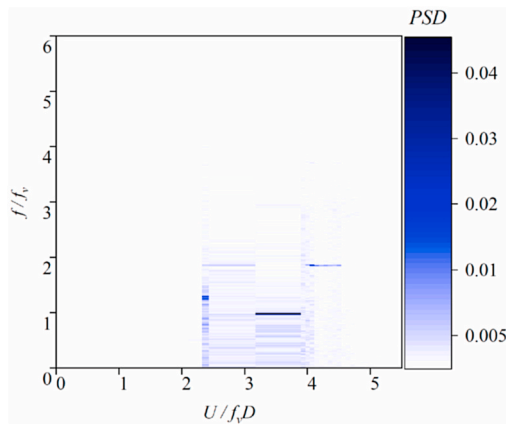


Fig. 10. PSD contour for the wake of case III.

It is worth mentioning that due to the strong separation at the leading edge of the central box, the separated vortex was so large that the KV in gap2 (KV-2) was spoiled by the downstream box.

In addition, another group of vortices was identified (Group 5).

Group 5 (KV-w): Due to the limitations of the experimental site, we could not obtain a smoke-line image of the wake. However, according to Wang et al. (2022) and Yang et al. (2021a), there should be interactions between double shear layers in the wake and thus the presence of Kármán vortices is expected.

Fig. 12 shows the flow dynamics for case I. When gap2 was covered, the influence of the vortices in Group 4 (MIV-3) was eliminated, i.e., the MIVs were not strengthened. Thus, unlike in cases 0 and II, the VIV

amplitude did not increase sharply around $U/fD = 4.60$ as shown in Fig. 5. Consequently, gap2 was essential for the further exacerbation of the VIV at $U/fD = 4.60$.

As discussed by Matsumoto et al. (1993), the interactive effects of KVs and MIVs depend on the geometric shapes of girder sections. Morikawa et al. (1991) found that the KV-spoiling effect of the downstream girder strengthened MIVs and thus increased the amplitude of VIVs in the upstream girder. Similarly, for the original girder in the present study, the KV-spoiling effect of the downstream girder eliminated KV-2. As a result, MIV-3 was strengthened and the VIV was strengthened as well.

Fig. 13 shows the flow dynamics for case II. When gap1 was covered, the influence of the vortices in Groups 2 and 3 (KV-1 and MIV-2) was eliminated. Small-scale MIV-1 separated from the leading edge of the upstream box and propagated downstream to the trailing edge of the central box. Due to the decrease in the scale of the separated vortices, KVs appeared in gap2 (KV-2). As shown in Fig. 5, the trend of the VIV amplitude variation looked the same as that of case 0. However, the VIV amplitude values decreased significantly in the whole VIV process. The decrease in the scale of MIVs coupled the decrease in the VIV amplitude, which cannot help decide the role of MIV-2. It can be inferred that the existence of gap1 is critical to the increase in the scale of MIVs as well as the VIV amplitude in the whole VIV process.

Fig. 14 visualizes the flow dynamics in case III. With all gaps covered, the model was a streamlined single-box girder and only MIV-1 and KV-w remained. VIVs did not occur and the MIV-1s were eliminated as well. The inflow separated at the leading edge and rolled up. In general, due to the Kelvin-Helmholtz (K-H) instability of the separated boundary layer, it would be expected that the separated shear layers would be shed from the tail, then would propagate downstream, merge, and produce alternative vortex shedding (KV-w) in the wake region. A typical example of

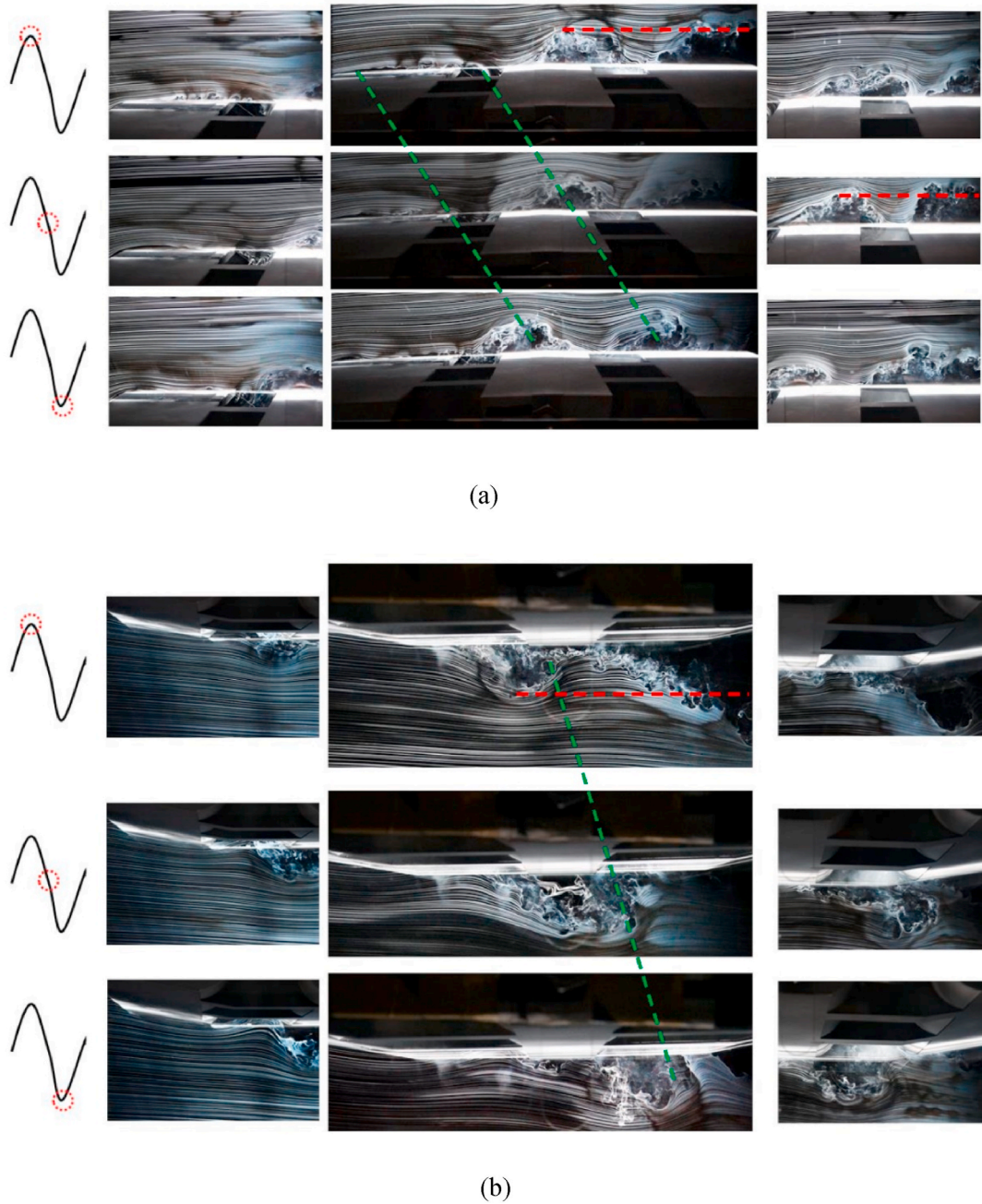


Fig. 11. Smoke-wire flow visualization for the vibrating bare girder when the VIV reached the maximum. (a) Upper surface; (b) Lower surface.

this process is the separation and wake flows of a 1:5 rectangular cylinder. However, as shown in Fig. 14, this rolling and merging of the separated shear layers did not happen in the near wake, mainly due to the streamline-shaped section of the girder model. Therefore, we could not obtain the domain wake frequency (Fig. 10). Thus, it is clear that the existence of gaps is essential for the formation of VIVs in a triple-box girder.

3.1.4. Discussion

The vortices in Groups 1–5 were present throughout the VIV process. However, there were dominant vortex types in certain VIV regions. As explained above, MIV-3 should be responsible for the sharp increase in the VIV amplitude and should dominate at the peak region. In this section, we will discuss the dominant regions of the other vortex types.

As we obtained $St_1 = 0.4$ for the original girder at rest, the KV-1-

induced VIV onset reduced velocity could be estimated as:

$$\frac{U}{f_v D} = \frac{1}{St}, \quad (1)$$

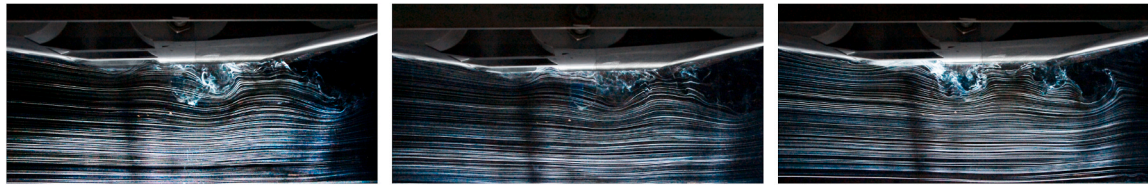
where U was the inflow velocity and f_v and D were the natural vertical frequency and the height of the test model, respectively. We could then calculate $U_{KV-1}/f_v D_1 = 2.5$, corresponding to the VIV onset wind velocity in Fig. 5. Thus, KV-1 dominated the VIV 'Beginning' region. However, in most regions of the VIV process, MIV coupling with the structural vibrations played a dominant role and sustained the VIVs.

3.1.5. Effects of the angle of attack

Fig. 15 illustrates the effects of the AOA on the VIV responses. As the AOA increased from -5° to 0° , the VIV amplitude increased monotonically and the onset wind velocity increased slightly. As the AOA



(a)



(b)

Fig. 12. Smoke-wire flow visualization for case I when the VIV reached the maximum. (a) Upper surface; (b) Lower surface.



(a)



(b)

Fig. 13. Smoke-wire flow visualization for case II when the VIV reached the maximum. (a) Upper surface; (b) Lower surface.

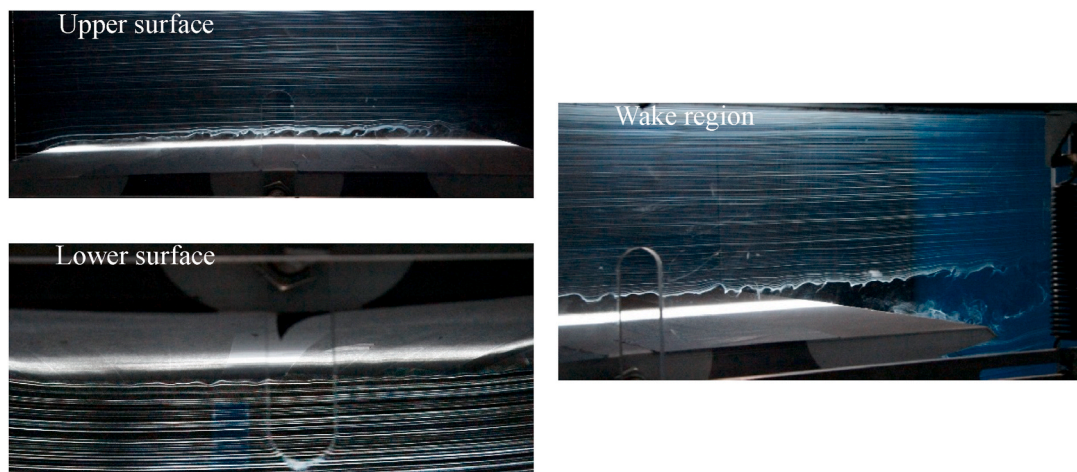


Fig. 14. Smoke-wire flow visualization for case III.

increased from 0° to 5° , the onset wind velocity kept increasing. The VIV amplitude was eventually saturated, but the VIV region was enlarged.

3.2. The girder with accessory structures

3.2.1. The effects of various combinations of accessory structures

Fig. 16 shows the VIV amplitudes of the triple-box girder with

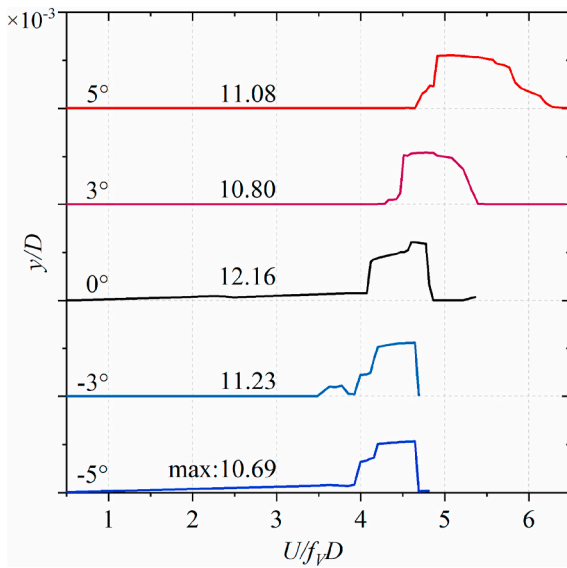


Fig. 15. VIV responses of the bare girder at various AOAs.

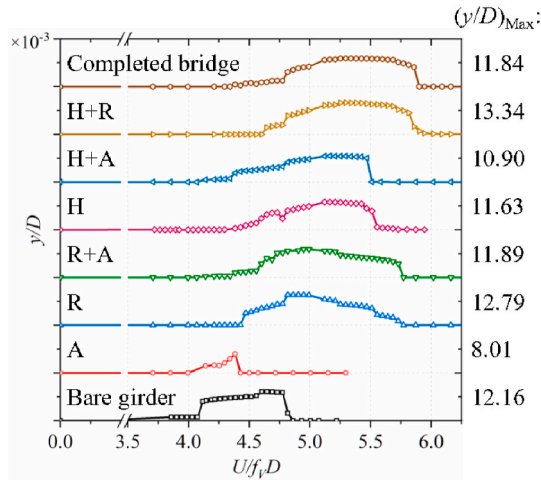


Fig. 16. VIV responses of the bare girder installed with various combinations of accessory structures.

various combinations of accessory structures. Compared to the bare girder, 'H' or 'R' increased the onset wind velocity of the VIV and enlarged the lock-in region. 'A' decreased the lock-in region as well as the VIV amplitude. This difference may be related to the locations of the accessory structures. The highway wind barrier and the railing at the upstream box-girder enhanced the VIV while the railway wind barrier at the central box-girder weakened it. 'H' + 'R' had the largest VIV amplitude, increased by about 9.7% over that of the bare girder. 'R' contributed more to the increase of VIV amplitude than 'H'. As long as the railway wind barrier was added, the VIV amplitude was reduced. Thus, to summarize, accessory components located at the upstream box-girder enhanced the VIVs while those at the central box-girder weakened them.

The present study could provide a reference for the installation of accessory structures on bare girders. The installation of a railway wind barrier should be given priority to effectively suppress VIVs during the installation of accessory structures.

3.2.2. Smoke-wire flow visualization

To demonstrate the coupling effects of the flow dynamics and the model motion, Fig. 17 shows the smoke-wire flow visualization of the

completed bridge with accessory structures. As shown in Fig. 17(a), flows through the highway wind barrier became quite turbulent (like small-scale jets) and then dissipated. Meanwhile, the separation point moved to the windward highway wind barrier. Due to the dissipation of the inflow and the blockage effect of the railway wind barrier, the flows on the upper surfaces of the two downstream boxes were quite steady. Meanwhile, separated gap vortices could be observed in gap1, which were formed by the lower shear layer alone [Fig. 17(b)]. These separated vortices coupled with the structural motion of the girder and propagated downstream due to the effects of the outer shear layer.

4. Conclusions

The present study has investigated long-span bridge triple-box girder cross-sections using wind tunnel tests. The VIV responses and the surrounding flow dynamics were investigated to reveal the VIV mechanisms of the original girder. The effect of AOAs was also analyzed. Various combinations of accessory structures were investigated for practical engineering purposes. The specific conclusions are as follows:

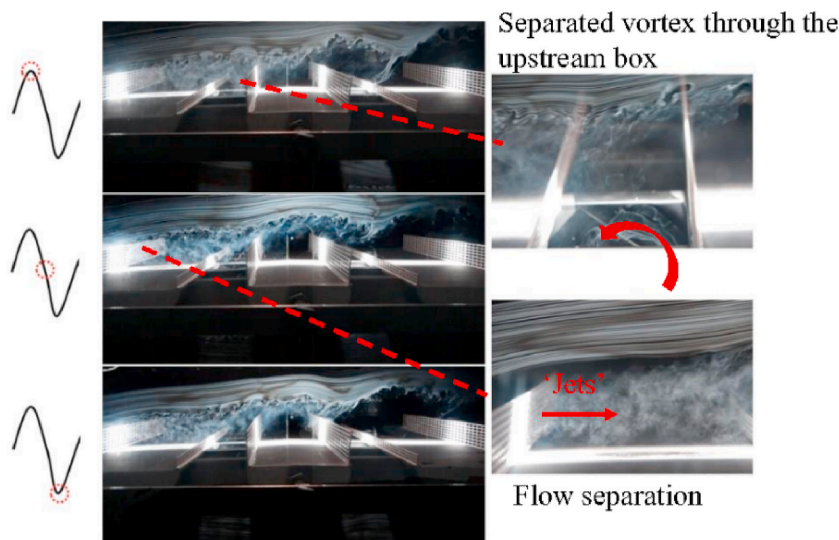
- (1) Under static conditions, a 'dual-frequency' phenomenon was seen in the two gaps of the bare girder. This phenomenon can enlarge the VIV region of the bridge under construction. Due to the streamline-shaped section, separated shear layers did not roll up and merge in the near wake, and as a result, no dominant frequency could be identified in the very near wake.
- (2) Gaps are essential for the occurrence of VIVs. Gap1 was essential for the enhancement of MIVs and thus the increase of VIV amplitude in the whole VIV process. The existence of gap2 led to the sudden exacerbation of VIV at a critical wind velocity. The VIV process could be classified into five stages: beginning, ascent 1, ascent 2, peak, and descent. For most VIV regions, MIVs coupled with the structural heaving motions to sustain the VIVs. There were five groups of vortex types: MIV-1, KV-1, MIV-2, MIV-3, and KV-w. It was verified that KV-1 dominated at the beginning and MIV-3 was crucial for the sudden exacerbation of VIV. The specific roles of MIV-1, MIV-2 and KV-w need further investigation.
- (3) The inflow angles of attack had significant effects on the VIV responses of the triple-box girder. The VIV onset wind velocity increased monotonically with the angle of attack. When the angle of attack was negative, the VIV amplitude increased with an increasing angle of attack.
- (4) Research on the installation of combinations of accessory structures was carried out. The railing contributed more to the increase of VIV amplitude than the highway wind barrier. The railway wind barrier was found to suppress the VIV amplitude and the VIV region. It is therefore recommended to install the railway wind barrier first, before a highway wind barrier and a railing.

Credit authorship contribution statement

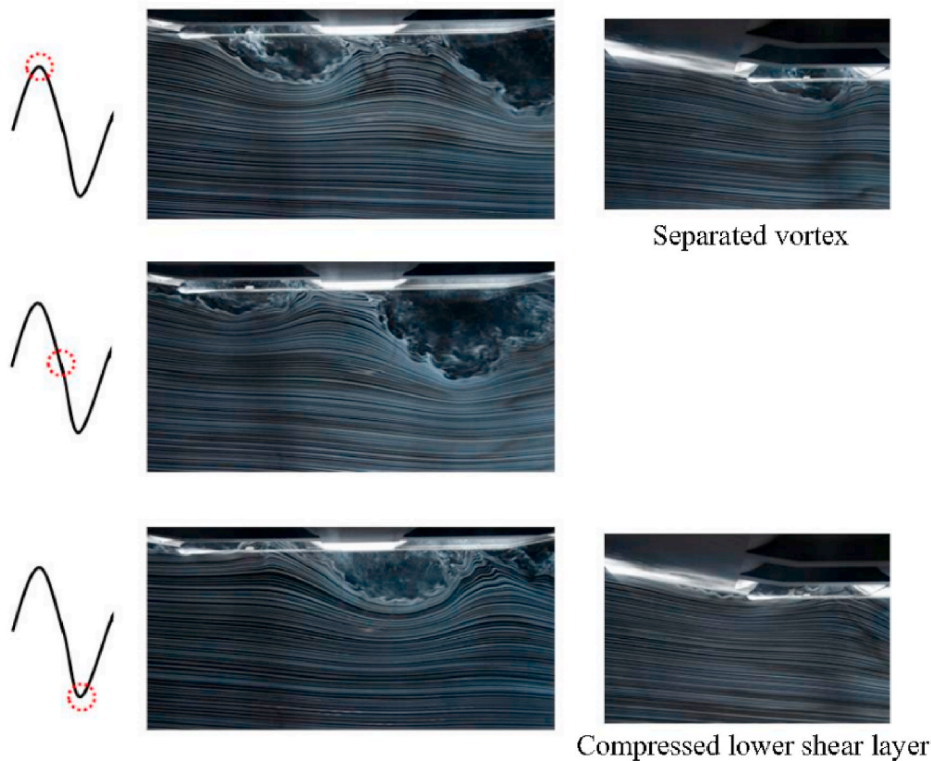
Hao Meng: experiments, data, figures, and writing of the draft manuscript. **Wen- Li Chen:** supervision, methodology, funding acquisition, and editing of the manuscript. **Hui Li:** supervision and editing of the manuscript. **Donglai Gao:** Conceptualization, methodology, experiments, data, figures, and writing of the manuscript.

Availability of data

The data that support the findings of this study are available from the corresponding author upon reasonable request.



(a) Upper surface



(b) Lower surface

(b) lower surface

Fig. 17. Smoke-wire flow visualization for the completed bridge while the VIV amplitude reached the largest.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

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